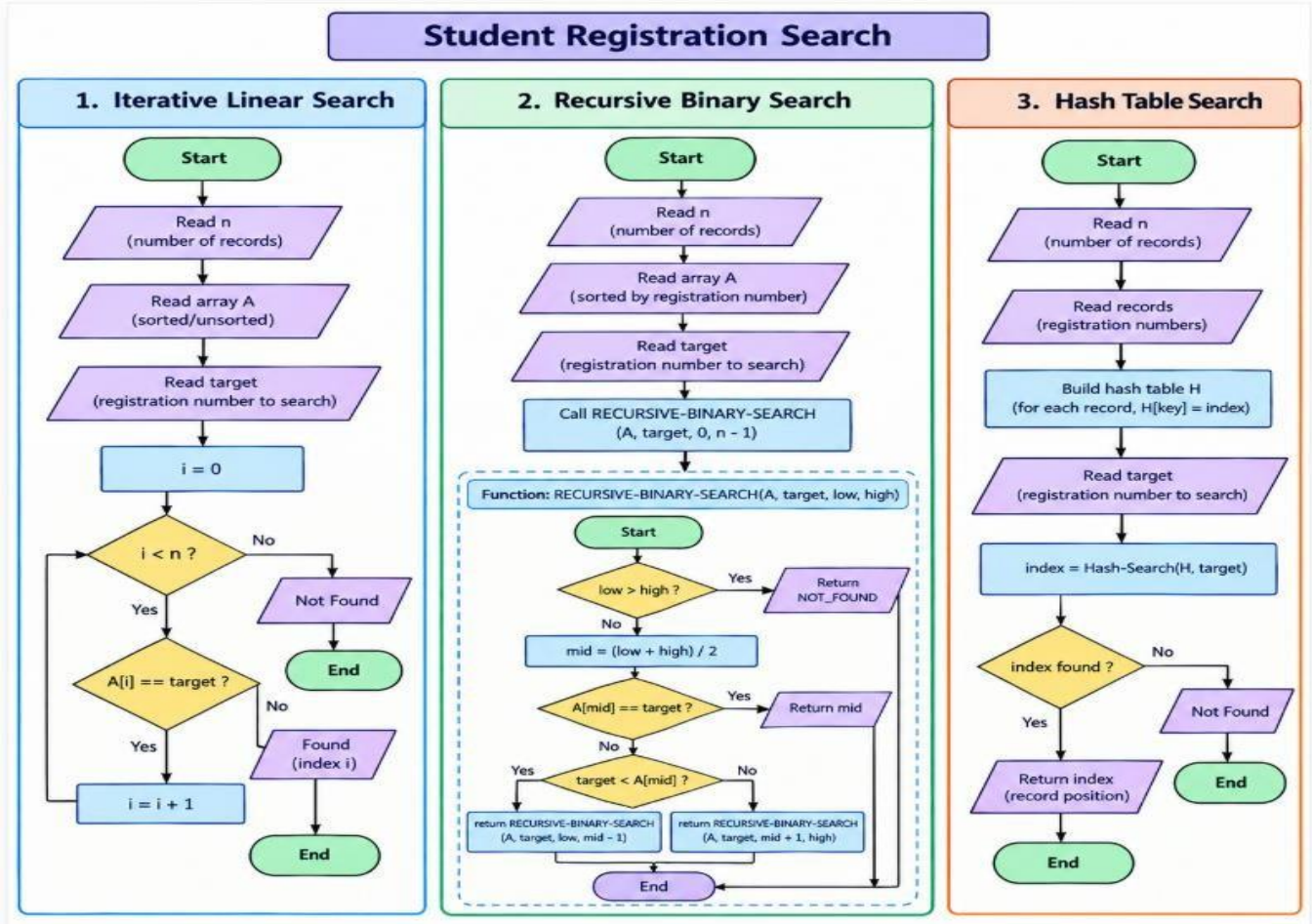
Task 8 – Implementation and testing

Language: Python

It generates the four specified dataset sizes, runs 30 successful random queries per algorithm/size, checks result correctness, tests missing keys, records mean and median elapsed milliseconds and comparison/lookup counts, and writes CSV plus graph PNG files. It uses matplotlib for graphs.

Flowchart



Test cases

Test	Input/Condition	Expected Result	Validation in Script
T1	Target is the first key in the sorted list.	Found at index 0.	Yes
T2	Target is the last key in the sorted list.	Found at index \(n- 1\).	Yes
T3	Target is present in the unsorted list.	Valid index and matching key returned.	Yes
T4	Target is absent (e.g., - 1, outside the generated range).	NOT_FOUND (index - 1).	Yes, if explicitly tested
T5	Random present targets across each dataset size.	Every search returns a matching key.	Yes